Help me understand

Radian Measure

Exam code: 9MA0

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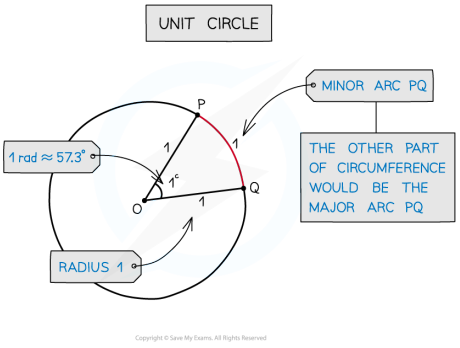
Radian measure

What are radians?

- **Radians** are an alternative to degrees for measuring angles



- 1 radian is the angle in a sector of radius 1 and arc length 1
- Radians are normally quoted in terms of π
- This leads to
  - **2π = 360°**
  - **π = 180°**
- The symbol for radians is ^{c} but it is more usual to see **rad**
  - Often, when π is involved, no symbol is given as it is obviously in radians

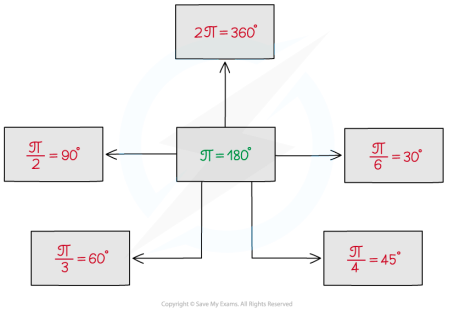
Stuck on something?

Ask questions and get help with exactly what you're revising.

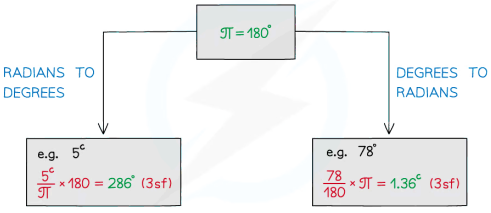
Got it

How do I convert between radians and degrees?

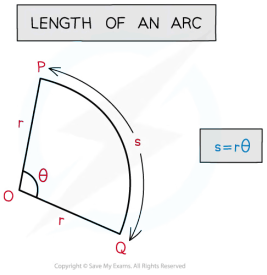
- The most common are below, these are helpful to know



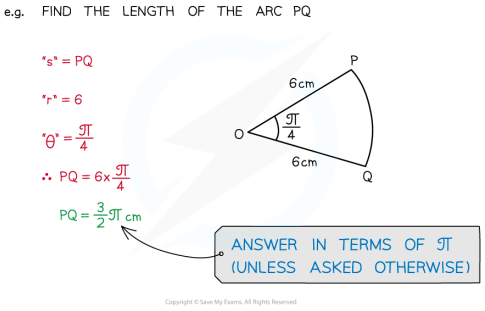
- Multiples of these are quite common too, eg $\frac{3}{4}\pi = 135°$
- For less familiar angles use **$\pi = 180°$** to convert

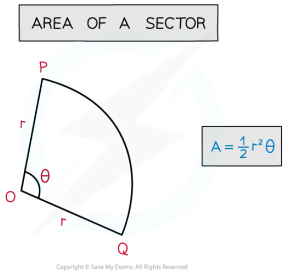


How do I find the arc length of a sector with radians?

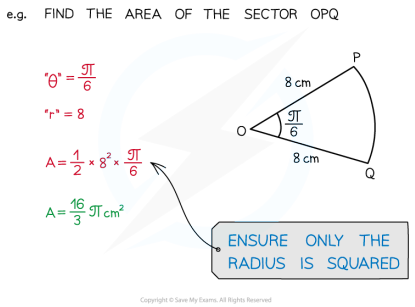


- The length of an arc is **s = rθ**



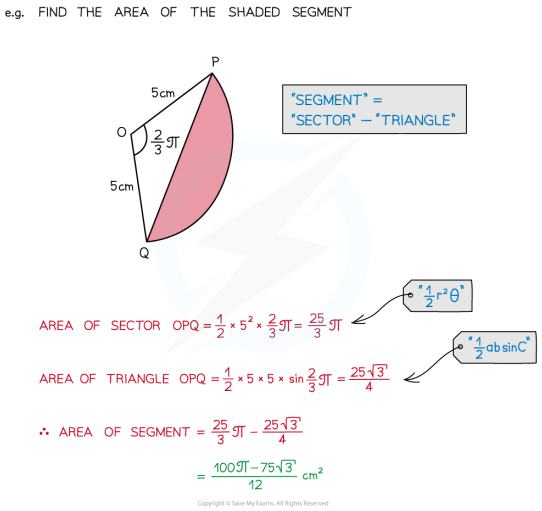


- The area of a sector is **A = ½r²θ**



How do I solve sector problems with radians?

- Other **angle**, **circle**, **area**, etc skills may be needed
- For example, area of a triangle **A = ½absinC**



Examiner Tips and Tricks

- Exam papers will have a mixture of questions in degrees and radians

calculator

- For the Casio fx-991EX Classwiz
  - **SHIFT** then **MENU SETUP**
  - Choose option **2: ANGLE UNIT**
  - Choose either **1: DEGREE** or **2: RADIAN**

Worked Example

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